

# Smart Home AIoT – Test Data Collection Sheet – Phase G – Schedule Automations

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Instructions for use:

1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:

- Pass Rate =  $(\text{Pass Count} / 10) \times 100\%$
- Avg RT =  $(\text{response times}) / 10$
- Variance =  $(x_i - )^2 / n$
- Std Dev =  $\sqrt{\text{variance}}$

Coverage Matrix

| Phase   | Description                    | Count |
|---------|--------------------------------|-------|
| Phase G | Phase G – Schedule Automations | 7     |
|         | TOTAL                          | 7     |

## Phase G – Schedule Automations

| G01                                    | Daily Watering 07:00 (10s) |                  |               |                 | Phase G |                  |       |
|----------------------------------------|----------------------------|------------------|---------------|-----------------|---------|------------------|-------|
| Method:     wait or force time = 07:00 |                            |                  |               |                 |         |                  |       |
| Expected:   pump turn_on 10s turn_off  |                            |                  |               |                 |         |                  |       |
| #                                      | Date                       | Time             | Input/Command | Actual Output   | RT (ms) | P/F              | Notes |
| 1                                      |                            |                  |               |                 |         |                  |       |
| 2                                      |                            |                  |               |                 |         |                  |       |
| 3                                      |                            |                  |               |                 |         |                  |       |
| 4                                      |                            |                  |               |                 |         |                  |       |
| 5                                      |                            |                  |               |                 |         |                  |       |
| 6                                      |                            |                  |               |                 |         |                  |       |
| 7                                      |                            |                  |               |                 |         |                  |       |
| 8                                      |                            |                  |               |                 |         |                  |       |
| 9                                      |                            |                  |               |                 |         |                  |       |
| 10                                     |                            |                  |               |                 |         |                  |       |
| Pass Count: ____/10                    |                            | Pass Rate: ____% |               | Avg RT: ____ ms |         | Std Dev: ____ ms |       |
| Comments:                              |                            |                  |               |                 |         |                  |       |

G02

Lights OFF 23:00

Phase G

Method: wait or force time = 23:00

Expected: all lights OFF

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments:

Method: wait or force time = 06:00

Expected: Tuya gradual brighten

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments:

G04

Garden Light ON 18:00

Phase G

Method: wait or force time = 18:00

Expected: switch.garden\_light ON

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments:

G05

Garden Light OFF 23:00

Phase G

Method: wait or force time = 23:00

Expected: switch.garden\_light OFF

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments:

G06

Sunset Curtain Close

Phase G

Method: force sun.sun below\_horizon (Home mode)

Expected: bedroom\_curtain close

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments:

Method: force sun event + 30 min (not Sleep)

Expected: bedroom\_curtain open

| #  | Date | Time | Input/Command | Actual Output | RT (ms) | P/F | Notes |
|----|------|------|---------------|---------------|---------|-----|-------|
| 1  |      |      |               |               |         |     |       |
| 2  |      |      |               |               |         |     |       |
| 3  |      |      |               |               |         |     |       |
| 4  |      |      |               |               |         |     |       |
| 5  |      |      |               |               |         |     |       |
| 6  |      |      |               |               |         |     |       |
| 7  |      |      |               |               |         |     |       |
| 8  |      |      |               |               |         |     |       |
| 9  |      |      |               |               |         |     |       |
| 10 |      |      |               |               |         |     |       |

Pass Count: \_\_\_/10

Pass Rate: \_\_\_\_%

Avg RT: \_\_\_\_ ms

Std Dev: \_\_\_\_ ms

Comments: